1. What is the standard aspect ratio of length to width for ISO A-series drawing paper sheets?

(A) sqrt(2) : 1 (approximately 1.414 : 1)
(B) 4 : 3
(C) 16 : 9
(D) 2 : 1

2. Which standard organization governs engineering drawing conventions and practices in India?

(A) BIS (Bureau of Indian Standards)
(B) ISO (International Organization for Standardization)
(C) ANSI (American National Standards Institute)
(D) DIN (Deutsches Institut fur Normung)

3. According to ISO 216 and BIS SP 46, what are the untrimmed dimensions of an A0 sheet?

(A) 880 mm x 1230 mm
(B) 841 mm x 1189 mm
(C) 594 mm x 841 mm
(D) 420 mm x 594 mm

4. What are the trimmed dimensions of an A3 size engineering drawing sheet?

(A) 297 mm x 420 mm
(B) 210 mm x 297 mm
(C) 420 mm x 594 mm
(D) 148 mm x 210 mm

5. What is the surface area of a standard trimmed A0 drawing sheet?

(A) Exactly 1.0 square meter
(B) 0.5 square meter
(C) 2.0 square meters
(D) 1.5 square meters

6. Where is the title block permanently located on an engineering drawing sheet according to BIS SP 46?

(A) Bottom right-hand corner of the drawing space
(B) Top right-hand corner
(C) Bottom left-hand corner
(D) Centered at the bottom border

7. What is the standard recommended dimension of a title block according to BIS standards?

(A) 185 mm x 65 mm
(B) 150 mm x 50 mm
(C) 200 mm x 75 mm
(D) 170 mm x 60 mm

8. Which pencil lead hardness grade is best suited for drawing construction lines and projection lines?

(A) 2H to 4H
(B) HB to B
(C) 2B to 4B
(D) 6B to 8B

9. Which pencil lead grade is recommended for drawing final outlines, visible edges, and border lines?

(A) H or HB
(B) 4H or 5H
(C) 6H or 7H
(D) 3H or 4H

10. A T-square is primarily used in manual drafting for drawing which type of lines?

(A) Horizontal parallel lines
(B) Vertical parallel lines
(C) Inclined lines at 30 degrees
(D) Curved tangent contours

11. Which angles can be drawn directly by combining a 30-60 degree set-square and a 45-45 degree set-square?

(A) Multiples of 15 degrees
(B) Multiples of 10 degrees
(C) Multiples of 20 degrees
(D) Multiples of 25 degrees

12. What is the primary function of a Mini-Drafter in engineering drawing?

(A) Combines the functions of T-square, set-square, protractor, and scales
(B) Only measures line thickness
(C) Replaces French curves for irregular splines
(D) Magnifies drawing details

13. Which instrument is specifically designed to draw irregular curved lines that cannot be drawn with a compass?

(A) French curves (or flexible spline)
(B) Clinograph
(C) Proportional divider
(D) Beam compass

14. According to BIS SP 46:2003, which line type is used to represent visible outlines and visible edges?

(A) Continuous thick line (Type A)
(B) Continuous thin line (Type B)
(C) Dashed thin line (Type E)
(D) Chain thin line (Type G)

15. Which line type is designated for representing hidden outlines and hidden edges in engineering views?

(A) Dashed thin line (Type E or F)
(B) Continuous thick line (Type A)
(C) Chain thin line (Type G)
(D) Continuous thin freehand line (Type C)

16. What is the standard thickness ratio between thick lines and thin lines on an engineering drawing?

(A) 2 : 1
(B) 3 : 1
(C) 4 : 1
(D) 1.5 : 1

17. What is the recommended proportion of dash length to space in a standard hidden line?

(A) Dash 3 mm to 4 mm, space 1 mm
(B) Dash 10 mm, space 5 mm
(C) Dash 1 mm, space 1 mm
(D) Dash 6 mm, space 3 mm

18. Which line type is used for center lines, lines of symmetry, and pitch circles of gears?

(A) Long-dashed dotted thin line (Chain thin, Type G)
(B) Continuous thin line
(C) Dashed thick line
(D) Continuous zig-zag line

19. A continuous thin line with freehand waviness (Type C) is primarily used for:

(A) Limits of partial or interrupted views and sections
(B) Visible outer boundaries
(C) Cutting plane paths
(D) Center lines of shafts

20. What is the standard ratio of length to width for arrowheads used in engineering dimensioning?

(A) 3 : 1
(B) 2 : 1
(C) 4 : 1
(D) 1 : 1

21. In the aligned system of dimensioning, how are dimensions placed relative to the dimension line?

(A) Parallel to and placed above the dimension line
(B) Perpendicular to the dimension line
(C) Centered within an interruption in the dimension line
(D) At the ends of the extension lines

22. In the unidirectional system of dimensioning, how are dimensions oriented on the drawing sheet?

(A) Horizontally so they can be read from the bottom of the sheet only
(B) Vertically along the right edge
(C) Parallel to each respective dimension line
(D) At 45 degrees to the horizontal axis

23. What is the minimum distance an extension line should extend past the dimension line?

(A) 2 mm to 3 mm
(B) 5 mm to 8 mm
(C) 10 mm
(D) 0.5 mm

24. Which symbol is placed before a dimension figure to indicate the diameter of a circular feature?

(A) Symbol phi
(B) Symbol R
(C) Symbol S-phi
(D) Symbol CR

25. Which symbol precedes a dimension figure indicating a spherical radius?

(A) SR
(B) R
(C) phi
(D) S-phi

26. In single-stroke vertical gothic lettering, what does the term 'single stroke' signify?

(A) Line width is uniform and produced by a single pencil movement
(B) All letters must be drawn in one continuous stroke without lifting pencil
(C) Letters are drawn with double outline shading
(D) Each word is written as a single cursive string

27. What is the standard height-to-width ratio for most vertical capital letters (such as B, D, E, P, R)?

(A) 7 : 5
(B) 7 : 7
(C) 7 : 3
(D) 7 : 1

28. Which letter among the English capital alphabet has a standard width equal to its height (7:7)?

(A) W
(B) I
(C) M
(D) A

29. Which letter has the narrowest proportion with a width-to-height ratio of 1:7?

(A) I
(B) J
(C) L
(D) T

30. What is the mathematical definition of Representative Fraction (RF) of an engineering scale?

(A) Drawing length / Actual physical length (in identical units)
(B) Actual physical length / Drawing length
(C) Drawing area / Actual area
(D) Scale factor x Drawing sheet size

31. A scale with an RF of 1:1 is categorized as a:

(A) Full size scale
(B) Reducing scale
(C) Enlarging scale
(D) Comparative scale

32. Which of the following representative fractions represents an enlarging scale?

(A) 5 : 1
(B) 1 : 2
(C) 1 : 50
(D) 1 : 100

33. How many consecutive units of measurement can a plain scale represent?

(A) Two units (e.g., meters and decimeters)
(B) One unit only
(C) Three units (e.g., meters, decimeters, centimeters)
(D) Four units simultaneously

34. A diagonal scale is based on which fundamental geometric theorem?

(A) Principle of similar right-angled triangles
(B) Pythagorean theorem
(C) Thales theorem of intercepts
(D) Euler's formula for polyhedra

35. In AutoCAD, which command allows setting linear units, angular units, and numerical precision?

(A) UNITS
(B) LIMITS
(C) DIMSTYLE
(D) LTSCALE

36. Which AutoCAD command defines the boundaries of the grid display and drawing extent in model space?

(A) LIMITS
(B) BOUNDARY
(C) RECTANG
(D) MVSETUP

37. Which function key toggles Object Snap (OSNAP) ON and OFF in AutoCAD?

(A) F3
(B) F8
(C) F7
(D) F9

38. Which function key toggles Ortho mode ON and OFF to restrict cursor motion strictly horizontally and vertically?

(A) F8
(B) F3
(C) F2
(D) F10

39. Which function key toggles the grid display on the AutoCAD drawing canvas?

(A) F7
(B) F9
(C) F6
(D) F1

40. What is the command shortcut to activate the PAN tool in AutoCAD via keyboard?

(A) P (or pressing down the mouse wheel)
(B) Z
(C) M
(D) E

41. If an actual distance of 5 meters is represented by a line of 25 mm on an architectural plan, what is the RF?

(A) 1 / 200
(B) 1 / 50
(C) 1 / 20
(D) 1 / 500

42. What is the formula for calculating the Length of Scale (LOS) to be constructed on a drawing sheet?

(A) LOS = RF x Maximum Length to be measured (ML)
(B) LOS = RF / Maximum Length (ML)
(C) LOS = Maximum Length (ML) / RF
(D) LOS = RF x Total sheet length

43. A plain scale has an RF = 1:40 and is required to measure up to 6 meters. What is the length of scale (LOS)?

(A) 15 cm (150 mm)
(B) 12 cm (120 mm)
(C) 20 cm (200 mm)
(D) 24 cm (240 mm)

44. On a map, an area of 144 sq. cm represents an actual land area of 36 sq. km. What is the Representative Fraction (RF)?

(A) 1 / 50,000
(B) 1 / 25,000
(C) 1 / 100,000
(D) 1 / 500,000

45. Which type of scale is required when measurements must be read down to three units (e.g., meters, decimeters, centimeters)?

(A) Diagonal scale
(B) Plain scale
(C) Comparative scale
(D) Vernier scale

46. In the construction of a diagonal scale, how is a division of 1 mm further subdivided into 10 parts of 0.1 mm each?

(A) By drawing 10 equidistant parallel horizontal lines across a vertical height
(B) By dividing the line into 10 curved arcs
(C) By rotating the line through 45 degrees
(D) By magnifying the line under an optical lens

47. When dimensioning a circle in engineering drawing, how should the dimension line be positioned?

(A) Passing through the center of the circle inclined at an angle, with arrowheads touching the circumference
(B) Tangent to the outer diameter
(C) Parallel to the vertical edge of the sheet
(D) Along a chord of the circle

48. Which dimensioning arrangement avoids the progressive accumulation of manufacturing tolerances?

(A) Datum (parallel) dimensioning
(B) Chain (continuous) dimensioning
(C) Combined dimensioning
(D) Coordinate dimensioning without datum

49. When dimensions are placed in a continuous series end-to-end, this arrangement is called:

(A) Chain dimensioning
(B) Parallel dimensioning
(C) Running dimensioning
(D) Superimposed dimensioning

50. What is the standard gap maintained between the visible outline of an object and the start of an extension line?

(A) 1 mm to 1.5 mm
(B) 5 mm to 6 mm
(C) 0 mm (touching the object outline)
(D) 10 mm

51. Why should dimension lines never cross extension lines or other dimension lines whenever possible?

(A) To prevent visual confusion and misinterpretation of engineering drawings
(B) Because the drawing ink will smudge
(C) Because AutoCAD generates an automatic error
(D) To save paper space

52. In AutoCAD, what does typing '@50<45' at the command prompt signify?

(A) Relative polar coordinate: distance 50 units at an angle of 45 degrees from last point
(B) Absolute coordinate (50, 45)
(C) Relative cartesian coordinate: dx = 50, dy = 45
(D) A point 50 units along Z axis

53. In AutoCAD, what does typing '@40,-25' specify during entity creation?

(A) Relative Cartesian displacement: +40 units in X and -25 units in Y from previous point
(B) Absolute point at X=40, Y=-25
(C) Polar coordinate of length 40 at -25 degrees
(D) A vector with magnitude sqrt(40^2 + 25^2)

54. What is the difference between WCS (World Coordinate System) and UCS (User Coordinate System) in AutoCAD?

(A) WCS is the fixed global reference frame, while UCS can be relocated and rotated by the user
(B) WCS is 2D and UCS is strictly 3D
(C) WCS uses inches and UCS uses millimeters
(D) WCS is for printing and UCS is for drafting

55. Which AutoCAD command allows the user to reorient the User Coordinate System by specifying three points?

(A) UCS -> 3point (Origin, Positive X-axis, Positive Y-axis)
(B) ROTATE
(C) ALIGN
(D) COORDINATE

56. Which command in AutoCAD controls the visibility and appearance of the UCS icon?

(A) UCSICON
(B) ICONMODE
(C) DISPLAYUCS
(D) VIEWUCS

57. What is the function of the ZOOM -> ALL command in AutoCAD?

(A) Displays the entire drawing limits or the full extent of drawn objects, whichever is larger
(B) Zooms in by a factor of 2x
(C) Zooms to a selected single object
(D) Resets screen brightness

58. What does ZOOM -> EXTENTS do in AutoCAD?

(A) Calculates the zoom factor to display all drawn objects filling the current viewport
(B) Resets drawing to the origin (0,0)
(C) Zooms to predefined grid limits only
(D) Restores the previous view

59. Which Object Snap (OSNAP) mode accurately snaps to the four cardinal points (0, 90, 180, 270 deg) of a circle or arc?

(A) Quadrant
(B) Center
(C) Tangent
(D) Node

60. Which OSNAP mode snaps to a point along an object that forms a 90-degree angle with another entity?

(A) Perpendicular
(B) Tangent
(C) Normal
(D) Orthogonal

61. Which OSNAP mode snaps to point objects, dimension definition points, or hatch origins?

(A) Node
(B) Endpoint
(C) Center
(D) Insertion

62. What is the role of Object Snap Tracking (F11) in AutoCAD?

(A) Enables temporary tracking vector paths aligned with acquired OSNAP points
(B) Locks cursor to absolute grid nodes
(C) Restricts drafting to isometric planes
(D) Automatically converts splines to arcs

63. What does Dynamic Input (F12) provide when active in AutoCAD?

(A) Displays command prompts, cursor coordinates, and dimensional input fields near the crosshairs
(B) Automatically saves files every 5 minutes
(C) Animates 3D camera orbits
(D) Predicts the next drawing command using AI

64. Which function key toggles Polar Tracking ON and OFF in AutoCAD?

(A) F10
(B) F8
(C) F3
(D) F11

65. In AutoCAD drafting, how does Polar Tracking differ from Ortho mode?

(A) Polar tracking allows custom incremental angle tracking (e.g. 15, 30, 45 deg), while Ortho locks strictly to 90 deg increments
(B) Polar tracking works only in 3D views
(C) Ortho allows freehand curved lines
(D) Polar tracking disables OSNAP

66. What is the standard order of line precedence when two or more lines coincide on an engineering drawing?

(A) Visible outline > Hidden line > Cutting plane line > Centerline
(B) Centerline > Visible outline > Hidden line > Cutting plane line
(C) Hidden line > Visible outline > Centerline > Cutting plane line
(D) Cutting plane line > Centerline > Hidden line > Visible outline

67. When a center line terminates, how far should it extend beyond the visible outline of the feature?

(A) 2 mm to 5 mm
(B) 10 mm to 15 mm
(C) It must terminate exactly on the outline
(D) At least 20 mm

68. Why should hidden lines never be drawn directly through the center of a visible outline?

(A) To maintain clarity and prevent the hidden line from being misinterpreted as a centerline or visible edge
(B) Because it is prohibited by CAD software
(C) Because hidden lines cannot be printed
(D) To reduce plotter file size

69. In single stroke vertical lettering, what is the nominal height of standard dimension numerals on A3 drawings?

(A) 3.5 mm
(B) 1.5 mm
(C) 7.0 mm
(D) 10.0 mm

70. What is the standard recommended height for main drawing titles in the title block of an A3 sheet?

(A) 7 mm or 10 mm
(B) 2.5 mm
(C) 3.5 mm
(D) 20 mm

71. What is the recommended spacing between two consecutive words in technical lettering?

(A) Equal to the width of the capital letter 'M' (or approximately 0.6 times the letter height h)
(B) Equal to 2 mm regardless of height
(C) Equal to the full sheet margin
(D) One millimeter

72. In AutoCAD, what does typing 'LIMITS 0,0 420,297' configure?

(A) Sets the lower-left corner to (0,0) and upper-right corner to (420,297) for standard A3 drawing sheet
(B) Draws a rectangle of 420x297 mm
(C) Cuts the viewport to 420x297 pixels
(D) Defines the plot scale as 1:420

73. When drawing a plain scale, where is the zero mark ('0') placed on the graduated line?

(A) At the end of the first main division, separating the main unit from the sub-units
(B) At the extreme left edge of the scale
(C) At the extreme right edge of the scale
(D) Exactly at the center of the scale

74. In which direction are sub-units numbered from the zero ('0') mark on a plain scale?

(A) Numbered towards the left from 0
(B) Numbered towards the right from 0
(C) Numbered upwards vertically
(D) Numbered in descending order from the right end

75. What is the purpose of the Ebony edge on a traditional wooden drawing board?

(A) Provides a perfectly straight, smooth reference edge for the stock of the T-square to slide along
(B) Prevents the board from warping in humidity
(C) Serves as an integrated millimeter ruler
(D) Protects the pencil lead from breaking

76. A room of actual volume 80 cubic meters is represented by a model volume of 10,000 cubic centimeters. What is the Representative Fraction (RF)?

(A) 1 / 20
(B) 1 / 40
(C) 1 / 8
(D) 1 / 200

77. A distance of 40 km between two railway stations is represented by 10 cm on a survey map. What is the length of scale needed to measure up to 50 km?

(A) 12.5 cm
(B) 15.0 cm
(C) 10.0 cm
(D) 20.0 cm

78. A diagonal scale is designed with RF = 1:2.5 to show decimeters, centimeters, and millimeters, with a maximum length of 4 dm. What is the length of scale (LOS)?

(A) 16 cm (160 mm)
(B) 10 cm (100 mm)
(C) 20 cm (200 mm)
(D) 25 cm (250 mm)

79. In the diagonal scale of Q78, if the vertical height is divided into 10 equal parts to read millimeters, what is the value represented by each vertical division?

(A) 1 mm (0.1 cm or 0.01 dm)
(B) 0.1 mm
(C) 0.5 mm
(D) 2 mm

80. A rectangular plot of land measuring 16 sq. km is shown on a map by a rectangle of 64 sq. cm. What is the scale of the map and its RF?

(A) 1 cm = 500 m (RF = 1 / 50,000)
(B) 1 cm = 250 m (RF = 1 / 25,000)
(C) 1 cm = 1 km (RF = 1 / 100,000)
(D) 1 cm = 2 km (RF = 1 / 200,000)

81. Using the map scale from Q80 (RF = 1:50,000), what is the length of scale required to measure up to a maximum distance of 7 km?

(A) 14 cm (140 mm)
(B) 12 cm (120 mm)
(C) 15 cm (150 mm)
(D) 10 cm (100 mm)

82. A line segment drawn from point P(15, 30) using relative polar coordinates '@80<135' terminates at point Q. What are the absolute Cartesian coordinates of Q?

(A) (-41.57, 86.57)
(B) (71.57, 86.57)
(C) (-41.57, -26.57)
(D) (95.00, 110.00)

83. In AutoCAD, three points are given: A(10, 20), B(50, 20), and C(50, 60). What is the area of the polygon formed by connecting A -> B -> C -> A using the AREA command?

(A) 800 square units
(B) 1600 square units
(C) 400 square units
(D) 1200 square units

84. If the drawing limits in AutoCAD are set to (0,0) and (420,297), and 'LIMITS -> ON' is executed, what happens when a user attempts to draw a line starting at (-10, 50)?

(A) AutoCAD rejects the point and displays message '**Point outside limits'
(B) The point is accepted and limits expand automatically
(C) The drawing file is corrupted
(D) AutoCAD crashes immediately

85. A vernier scale has 10 divisions on the vernier plate equal in length to 9 divisions of the primary main scale (each main scale division = 1 mm). What is the least count?

(A) 0.1 mm (Direct Vernier)
(B) 0.01 mm
(C) 0.05 mm
(D) 0.2 mm

86. In a retrograde vernier scale, 10 vernier divisions are equal in length to how many main scale divisions?

(A) 11 main scale divisions
(B) 9 main scale divisions
(C) 10 main scale divisions
(D) 12 main scale divisions

87. What is the mathematical condition for the least count (LC) of a forward (direct) vernier scale with main division 's' and vernier division 'v'?

(A) LC = s - v = s / n (where n is the number of vernier divisions)
(B) LC = v - s = v / n
(C) LC = s + v
(D) LC = s * v / n

88. A scale of chords is an engineering scale primarily constructed and used for:

(A) Measuring and laying out angles without using a protractor
(B) Measuring chord lengths of circular arches only
(C) Calculating gear tooth thickness
(D) Subdividing chords of ellipses

89. In constructing a scale of chords, the chord length C subtending an angle theta at the center of a circle of radius R is given by:

(A) C = 2 * R * sin(theta / 2)
(B) C = R * sin(theta)
(C) C = 2 * R * cos(theta / 2)
(D) C = R * tan(theta / 2)

90. In AutoCAD, what is the precise function of the LTSCALE system variable?

(A) Sets the global linetype scale factor for all dashed, hidden, and chain lines in the drawing
(B) Changes the font size of dimension text
(C) Scales the entire model geometry
(D) Adjusts the line weight display on monitor

91. If hidden lines appear as continuous solid lines in AutoCAD model space, which command adjustment resolves this display issue?

(A) Increase or decrease LTSCALE (and run REGEN)
(B) Change the screen resolution
(C) Use EXPLODE on the lines
(D) Turn on ORTHO mode

92. What does the command 'DIMSTYLE' control in an AutoCAD engineering drawing project?

(A) Dimension geometry, arrow styles, text placement, tolerance formats, and scale factors
(B) Only the font style of text notes
(C) The 3D rendering material of parts
(D) The layer color palette

93. In AutoCAD, if a user specifies an angle of -60 degrees, which standard positive angle is geometrically equivalent under default settings?

(A) 300 degrees
(B) 120 degrees
(C) 240 degrees
(D) 60 degrees

94. When drawing an engineering title block, what information is legally and technically required by ISO 7200 and BIS SP 46?

(A) Title, Drawing Number, Scale, Projection Symbol, Firm Name, Date, Author and Approver Signatures
(B) Only the author's personal phone number and date
(C) Cost of raw material and machine run time
(D) Only the software license serial number

95. Which geometric construction accurately divides an 85 mm straight line into 7 equal parts without numerical measurement error?

(A) Drawing an auxiliary line at an acute angle, stepping off 7 equal arcs, and drawing parallel lines to the end point
(B) Using an electronic digital caliper
(C) Trial and error with pencil marks
(D) Dividing 85 by 7 on a calculator and approximating to 12.14 mm with a ruler

96. A surveyor's chain of 20 meters is found to be 10 cm too long after measuring a distance recorded as 1500 meters. What is the true physical distance?

(A) 1507.5 meters
(B) 1492.5 meters
(C) 1500.0 meters
(D) 1515.0 meters

97. In AutoCAD, which option under the UNITS command configures angles to measure clockwise rather than the mathematical default counterclockwise?

(A) Check the 'Clockwise' checkbox in Drawing Units dialog
(B) Set ANGBASE to 180
(C) Set INSUNITS to 0
(D) Set UCS to View

98. What is the standard base angle orientation (0 degrees) in AutoCAD's default World Coordinate System?

(A) East (positive X-axis, pointing horizontally to the right)
(B) North (positive Y-axis, pointing straight up)
(C) West (negative X-axis)
(D) South (negative Y-axis)

99. What is the purpose of the 'Object Snap Tracking' (OSTRACK) alignment path when creating complex mechanical sketches?

(A) Enables precise construction of points based on orthogonal alignment with multiple reference snap points without drawing construction lines
(B) Snaps cursor strictly to drawing grid dots
(C) Prevents mouse movement outside viewport
(D) Automatically meshes 3D solids

100. In technical lettering, if the nominal letter height is h = 14 mm, what should be the standard minimum spacing between baselines (line spacing) as per BIS SP 46?

(A) 20 mm (approx 1.43 h)
(B) 14 mm (1.0 h)
(C) 10 mm (0.7 h)
(D) 28 mm (2.0 h)

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Explain the conceptual framework of engineering drawing as the universal language of engineers, detailing the role of national (BIS) and international (ISO) standards.

Q2. Classify and describe the standard drawing instruments used in manual drafting, including their proper maintenance and handling procedures.

Q3. Detail the standard sheet sizes of the ISO A-series (A0 to A5) based on ISO 216 and BIS SP 46, explaining their geometric relationship and trimming margins.

Q4. Explain the structure, standard dimensions, mandatory components, and placement of a Title Block in an engineering drawing according to BIS SP 46 and ISO 7200.

Q5. Categorize and describe the standard line types used in engineering drawing as per BIS SP 46:2003 (ISO 128), including their thicknesses, applications, and pencil selection.

Q6. Explain the fundamental principles and elements of dimensioning in technical drawings, contrasting the Aligned and Unidirectional dimensioning systems.

Q7. Discuss the rules for lettering in engineering drawing according to BIS SP 46 (ISO 3098), detailing single-stroke vertical gothic letters, heights, and spacing ratios.

Q8. Define Representative Fraction (RF), classify scales into full, reducing, and enlarging scales with engineering examples, and derive Length of Scale (LOS).

Q9. Construct a plain scale of RF = 1:40 to show meters and decimeters, capable of measuring up to 6 meters. Provide full mathematical calculations and step-by-step drafting procedure to mark a distance of 4.6 meters.

Q10. Design and construct a diagonal scale of RF = 1:5000 to show kilometers, hectometers, and decameters, suitable for reading up to 1 km. Provide all calculations and describe the procedure to indicate 6 hectometers and 7 decameters.

Q11. Explain the principle of diagonal scale construction based on the geometry of similar right triangles, deriving why it achieves higher measurement resolution.

Q12. Compare plain scales, diagonal scales, and vernier scales, summarizing their measurement capabilities, structural layout, and specific engineering applications.

Q13. Introduce the AutoCAD user interface, detailing its core components: Application Menu, Quick Access Toolbar, Ribbon, Drawing Area, Command Window, and Status Bar.

Q14. Explain the functionality of the UNITS command in AutoCAD, including length types, angular formats, precision settings, and insertion scale conventions.

Q15. Describe the LIMITS command in AutoCAD, explaining its role in setting drawing boundaries, how to enable/disable limit checking, and its relationship with ZOOM ALL.

Q16. Detail the navigation tools in AutoCAD: ZOOM options (All, Extents, Window, Dynamic), PAN, and mouse wheel shortcuts for viewport manipulation.

Q17. Explain the User Coordinate System (UCS) versus World Coordinate System (WCS) in AutoCAD, detailing the UCS command options and the UCSICON command.

Q18. Analyze the Object Snap (OSNAP) drafting system in AutoCAD, detailing all primary snap modes and their exact geometric triggering conditions.

Q19. Detail the Function Keys (F1 through F12) in AutoCAD, explaining their assigned drafting toggles and practical workflow significance.

Q20. Describe the coordinate entry systems in AutoCAD: Absolute Cartesian, Relative Cartesian, and Relative Polar, with step-by-step drafting examples.

Q21. Construct a diagonal scale with RF = 1:2.5 to show decimeters, centimeters, and millimeters, up to a maximum length of 4 decimeters. Show calculations and procedure to read 3.68 decimeters.

Q22. Explain the setup procedure for an AutoCAD drawing session for an A3 engineering sheet in millimeters, detailing all commands executed from scratch.

Q23. Provide the AutoCAD command sequence and prompt interactions to construct a standard A3 drawing border and a 185 mm x 65 mm title block with text.

Q24. Discuss the causes and remedies of common line appearance errors in AutoCAD, such as hidden and center lines appearing solid, detailing LTSCALE, PSLTSCALE, and REGEN.

Q25. Explain the methods of constructing an ellipse in engineering drawing and compare with AutoCAD's ELLIPSE command options (Axis End, Center, Isocircle).

Q26. Construct an engineering scale of chords to measure angles from 0 to 90 degrees in 5-degree increments. Derive the mathematical formula and describe how to construct an angle of 35 degrees using only a compass and scale of chords.

Q27. A map is drawn to a scale such that 1 cm represents 250 meters. Construct a comparative scale to read distances in miles and furlongs up to 4 miles. Provide complete mathematical conversions and drafting steps.

Q28. Explain the configuration and customization of Dimension Styles (DIMSTYLE) in AutoCAD, detailing primary settings for lines, symbols, arrows, text, fit, and primary units.

Q29. Explain the drafting methodology for dividing a circle into 12 equal parts using only standard set-squares and a compass, detailing its application in engineering projections.

Q30. Summarize the essential best practices for maintaining drawing integrity, clean CAD layer structure, file management, and quality assurance in professional engineering drafting.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[A]	**Q21:**	[A]	**Q41:**	[A]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[A]	**Q22:**	[A]	**Q42:**	[A]	**Q62:**	[A]	**Q82:**	[A]
Q3:	[A]	**Q23:**	[A]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[A]	**Q24:**	[A]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[A]
Q5:	[A]	**Q25:**	[A]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[A]	**Q26:**	[A]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[A]	**Q27:**	[A]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[A]	**Q28:**	[A]	**Q48:**	[A]	**Q68:**	[A]	**Q88:**	[A]
Q9:	[A]	**Q29:**	[A]	**Q49:**	[A]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[A]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[A]	**Q31:**	[A]	**Q51:**	[A]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[A]	**Q32:**	[A]	**Q52:**	[A]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[A]	**Q33:**	[A]	**Q53:**	[A]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[A]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[A]	**Q35:**	[A]	**Q55:**	[A]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[A]	**Q36:**	[A]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[A]	**Q37:**	[A]	**Q57:**	[A]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[A]	**Q38:**	[A]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[A]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[A]	**Q40:**	[A]	**Q60:**	[A]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

- Keep pencil leads sharpened with sandpaper pads (chisel point for straight lines, conical point for compass and lettering).
- Store mini-drafters with clamping screws relaxed to prevent permanent spring tension distortion.

Angle projection cone).
- Sheet Number & Total Sheets: Displayed as 'Sheet 1 of 3'.
- Date and Signatures: Columns for 'Drawn by', 'Checked by', and 'Approved by' with respective dates.
- Material & Tolerances: General dimensional tolerance callout (e.g. '+/- 0.1 mm') and surface finish symbol.

3. Draftsman Title Block Division Layout:
- Width is segmented: 185 mm total width partitioned into institutional identifier (65 mm), title and drawing number (75 mm), and projection/approval data (45 mm). Height 65 mm is divided into 6-8 standardized text rows.

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1. Core Elements of Dimensioning:
- Dimension Line: A thin continuous line (Type B) indicating the extent and direction of a dimension, terminated by arrowheads.
- Extension Line (Projection Line): Extends outwards from the object feature by 2 to 3 mm beyond the dimension line, leaving a gap of 1 mm from the feature contour.
- Arrowheads: Terminate dimension lines. Standard proportions are 3:1 (length:width), closed and solidly filled with black ink/lead.
- Leader Line: An inclined thin line leading from a note or dimension to a feature, terminating in an arrowhead (touching an outline) or dot (within a surface), with a horizontal landing of 2-3 mm at the note.

2. Comparison of Dimensioning Systems:
- Aligned System:
* Placement: Dimension values are placed parallel to the dimension line and positioned slightly above it (never touching the line).
* Reading Direction: Can be read from the bottom of the drawing sheet or from the right-hand edge.
* Application: Predominantly used in civil, architectural, and structural drawings.
- Unidirectional System:
* Placement: All dimension figures are oriented horizontally, breaking the dimension line near its center.
* Reading Direction: Read exclusively from the bottom of the drawing sheet, regardless of the angle of the feature.
* Application: Universally favored in mechanical engineering, aerospace, and CAD practice due to easier legibility on large sheets.

3. Golden Rules of Dimensioning:
- Never use centerlines or visible outlines as dimension lines.
- Place dimensions outside views wherever practical.
- Avoid redundant over-dimensioning; every feature must be dimensioned exactly once.

>>Solution to Question 7: Discuss the rules for lettering in engineering drawing according to BIS SP 46 (ISO 3098), detailing single-stroke vertical gothic letters, h

4. Standard Dimensional Proportions for Lettering Type A (d = h / 14):
- Height of capital letters: h (14/14 h).
- Height of lowercase letters (x-height): 10/14 h.
- Spacing between adjacent characters: 2/14 h.
- Spacing between baselines (line spacing): 20/14 h.
- Spacing between consecutive words: 6/14 h (width of letter 'M').
- Standard width ratio: 7:5 for capital letters B, D, E, F, G, H, K, N, O, P, Q, R, S, T, U, V, X, Y, Z; 7:7 for W; 7:6 for M; 7:1 for I.

Each subdivision represents 1 decimeter (0.1 m = 10 cm).
- Step 5: Number the decimeter marks from 0 towards the left: 0, 2, 4, 6, 8, 10 decimeters.
- Step 6: Print 'METERS' on the right, 'DECIMETERS' on the left, and 'RF = 1:40' below the scale.

3. Marking Target Distance of 4.6 Meters:
- Measure 4 meters from 0 towards the right (main scale mark 4).
- Measure 6 decimeters from 0 towards the left (subdivision mark 6).
- Draw extension lines vertically upward from mark 6 (left) and mark 4 (right).
- Draw a dimension line with arrowheads and label clearly '4.6 m'.

- Let ABC be a right-angled triangle where base BC has a unit length 's' (e.g., 1 mm or 1 sub-unit) and vertical altitude AB has arbitrary drafting height 'h' (e.g., 50 mm).
- Divide altitude AB into 'n' equal parts (typically n = 10) by horizontal lines parallel to base BC, labeled 1, 2, 3, ... n from top A down to base B.
- Let 1-1', 2-2', ... k-k' represent the horizontal lines at heights h1, h2, ... hk from apex A.
- By properties of similar triangles (Triangle A-k-k' is similar to Triangle A-B-C):
(k-k') / BC = (A-k) / AB = k / n.
Therefore: Length of horizontal segment (k-k') = (k / n) * BC = (k / n) * s.
- For n = 10:
* At 1st division from apex: length = 1/10 * s = 0.1 s.
* At 2nd division: length = 2/10 * s = 0.2 s.
* At 7th division: length = 7/10 * s = 0.7 s.

3. Resolution Advantage:
This geometric magnification converts a direct linear division of 0.1 mm (which is impossible to draw manually with a pencil) into a vertical step of 5 mm, achieving 10x greater drafting precision.

>> Solution to Question 10 Advance the AutoCAD user interface, delivering scalable, symmetrical, and efficient Quick Access Toolbar, Ribbon, Drawing Area

- Quick Access Toolbar (QAT): Customizable ribbon bar displaying common quick-action icons: Undo, Redo, Save, Plot, Workspace switcher.

2. Ribbon Interface:
- Organized hierarchically into Tabs (Home, Insert, Annotate, Parametric, View, Manage, Output) and Panels (Draw, Modify, Annotation, Layers, Block, Properties).
- Consolidates commands graphically with tooltips and expandable flyout menus.

3. Drawing Area (Graphic Canvas):
- The infinite virtual workspace where 2D and 3D geometric entities are created.
- Features the UCS Icon at bottom-left indicating active coordinate orientation, and the ViewCube at top-right for 3D viewpoint navigation.

4. Command Line Window:
- Located at the bottom of the workspace; the nerve center of AutoCAD.
- Displays real-time prompts, active sub-options (e.g., [Close/Undo]), and accepts typed command abbreviations. Pressing F2 expands the command history.

5. Status Bar:
- Docked at the very bottom right of the screen; contains toggle buttons for drafting aids: Model/Paper space, Grid (F7), Snap (F9), Ortho (F8), Polar (F10), OSNAP (F3), Dynamic Input (F12), Annotation Scale, and Workspace switching.

>>Solution to Question 14: Explain the functionality of the UNITS command in AutoCAD, including length types, angular formats, precision settings, and insertion s

Meters, Inches), enabling automatic unit conversion upon drag-and-drop.

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3. Standard Mouse Wheel Shortcuts:
- Roll Wheel Forward: Real-time zoom in centered at the mouse cursor location.
- Roll Wheel Backward: Real-time zoom out centered at mouse cursor.
- Press and Hold Wheel: Instantly activates Real-time Pan (hand cursor).
- Double-Click Wheel: Instantly executes ZOOM -> EXTENTS (fastest workflow in CAD practice).

Diamond).
- Intersection: Snaps to the intersection point of two lines, arcs, or curves (Marker: Green 'X').
- Extension: Displays a temporary extension tracking vector when hovering over an endpoint.
- Insertion: Snaps to the insertion base point of text, blocks, attributes, or shape files.
- Perpendicular: Snaps to a point on an entity that forms a 90? angle with the source point (Marker: Right-angle glyph).
- Tangent: Snaps to the tangent point on a circle, arc, or spline from the current point (Marker: Circle with tangent line).
- Node: Snaps to a point object (POINT command), hatch origin, or dimension definition point.
- Nearest: Snaps to the closest point on any visible entity contour (Marker: Hourglass).
- Apparent Intersection: Snaps to the visual intersection of two non-coplanar lines in 3D projection.

3. Best Practice:
Avoid keeping 'Nearest' permanently active in Running OSNAP as it overrides precision snaps like Endpoint or Midpoint.

>>Solution Question 28: The Function Keys (F1 through F12) in AutoCAD toggle Cartesian, Relative Cartesian, and Relative Polar coordinate display modes. Pressing F6 cycles through these modes, allowing the user to switch between absolute and relative coordinate entry methods during drafting.

- Example: To draw a line from origin to point (80, 60):
Command: LINE -> Specify first point: 0,0 [Enter] -> Specify next point: 80,60 [Enter].

2. Relative Cartesian Coordinates (@dX, dY):
- Syntax: @dX,dY (e.g. @50,-30)
- Reference: The '@' symbol instructs AutoCAD to measure displacement dX and dY relative to the last picked point.
- Example: From point (80,60), to move 40 units right and 25 units down:
Specify next point: @40,-25 [Enter] (Landing at absolute point 120, 35).

3. Relative Polar Coordinates (@Distance < Angle):
- Syntax: @distance<angle (e.g. @75<45)
- Reference: Specifies a straight vector of magnitude 'distance' inclined at 'angle' (measured counterclockwise from positive X-axis) relative to the last point.
- Example: To draw an equilateral triangle of side 100 mm starting at (0,0):
Command: LINE -> 0,0 [Enter]
Specify next point: @100<0 [Enter] (Base horizontal line)
Specify next point: @100<120 [Enter] (Inclined side at 120?)
Specify next point: C [Enter] (Closes back to 0,0).

>>Solution to Question 21: Construct a diagonal scale with RF = 1:2.5 to show decimeters, centimeters, and millimeters, up to a maximum length of 4 decimeters.

3. Set Border Lineweight:
- Command: PEDIT [Enter] -> Select rectangle -> Option: Width -> Enter: 0.5 [Enter].

4. Construct Title Block Outer Box (185 x 65 mm at bottom right):
- The bottom-right border corner is at (410, 10).
- Top-left corner of title block: X = 410 - 185 = 225; Y = 10 + 65 = 75.
- Command: RECTANG [Enter]
- Specify first corner point: 225,10 [Enter]
- Specify other corner point: 410,75 [Enter]

5. Subdivide Title Block Rows and Partitions:
- Command: EXPLODE [Enter] -> Select title block rectangle.
- Command: OFFSET [Enter] -> Specify distance: 15 [Enter] -> Offset top horizontal line downward by 15 mm.
- Command: OFFSET [Enter] -> Specify distance: 10 [Enter] -> Offset successive lines downward.
- Command: LINE [Enter] -> Draw vertical divider at X = 320 from Y = 10 to Y = 75.

6. Add Standard Text Notes:
- Command: TEXT [Enter] -> Justify: Middle Center (MC) -> Pick center of title cell.
- Specify height: 5.0 -> Rotation angle: 0 -> Text: 'LOVELY PROFESSIONAL UNIVERSITY' [Enter].
- Add 'SCALE: 1:1', 'PROJECTION: 1ST ANGLE', 'DWG NO: CSE-ED-01'.

>> **Solution to Question 24:** Discuss the causes and remedies of common line appearance errors in AutoCAD, such as hidden and center lines appearing solid, det

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(5?, 10?, 15? ... 90?), swing circular arcs down to intersect line AB.
- Step 5: Mark the divisions on line AB as 0?, 5?, 10?, 15? ... 90?. This straight graduated line is the Scale of Chords.

3. Constructing an Angle of 35? Using the Scale of Chords:
- Step 1: Draw a base reference ray PQ.
- Step 2: Set compass radius equal to chord 60? on the scale (which equals true reference radius R because 2*R*sin(30?) = R).
- Step 3: With P as center and radius R (chord 60?), swing a large circular arc intersecting ray PQ at point M.
- Step 4: From the scale of chords, take compass distance corresponding to 35?.
- Step 5: With M as center and compass radius = chord 35?, cut the circular arc at point N.
- Step 6: Draw ray PN through N. Angle MPN is precisely 35.0 degrees.

>>Solution to Question 25: Maintain drawing scale and customization of Sim 250 Series (DIMSTYLE) in AutoCAD, detailing dimensions, tolerances, and symbols

tolerances, and placement rules for dimensions, ensuring conformance with drafting standards.

2. Key Configuration Tabs:
- Lines Tab:
* Dimension Lines: Color (ByLayer), Linetype, Lineweight (0.25 mm), Baseline spacing (8-10 mm).
* Extension Lines: Extension beyond dimension line (set to 2-3 mm), Offset from origin (set to 1-1.5 mm).
- Symbols and Arrows Tab:
* Arrowheads: Set First, Second, and Leader to 'Closed filled'.
* Arrow size: 3.0 mm to 3.5 mm.
* Center marks: Set to 'Mark' or 'Line' (size 2.5 mm).
- Text Tab:
* Text Style: Standard font (ROMANS or ARIAL).
* Text Height: 3.5 mm for A3 sheets.
* Text Placement: Vertical = 'Above' (for Aligned system) or 'Centered' (for Unidirectional).
* Text Alignment: 'Aligned with dimension line' or 'Horizontal' (ISO standard).
- Fit Tab:
* Fit Options: 'Either text or arrows (best fit)' when space is tight between extension lines.
* Use overall scale of: 1.0 (for 1:1 drawing in model space).
- Primary Units Tab:
* Unit format: Decimal | Precision: 0.00 | Decimal separator: Period (.).
* Zero Suppression: Suppress trailing zeros (e.g. 15.00 becomes 15) or leading zeros as required.

>> Solution to Question 29: Explain the drafting methodology for dividing a circle into 12 equal parts using only standard set-squares and a compass, detailing its a

>>Solution to Question 30: Summarize the essential best practices for maintaining drawing integrity, clean CAD layer structure, file management, and quality assu